
Introduction to Machine Learning
(GAI 601)
WEEK 9
CU
NY
SPS

CUNY SPS Data Science Seminar

6:30PM Tuesday, November 3rd

CUNY@Amazon, Lobby 12 West 39th St. NY, NY 10018

Prof. Christian Martinez, CUNY/New York Open Data Lab

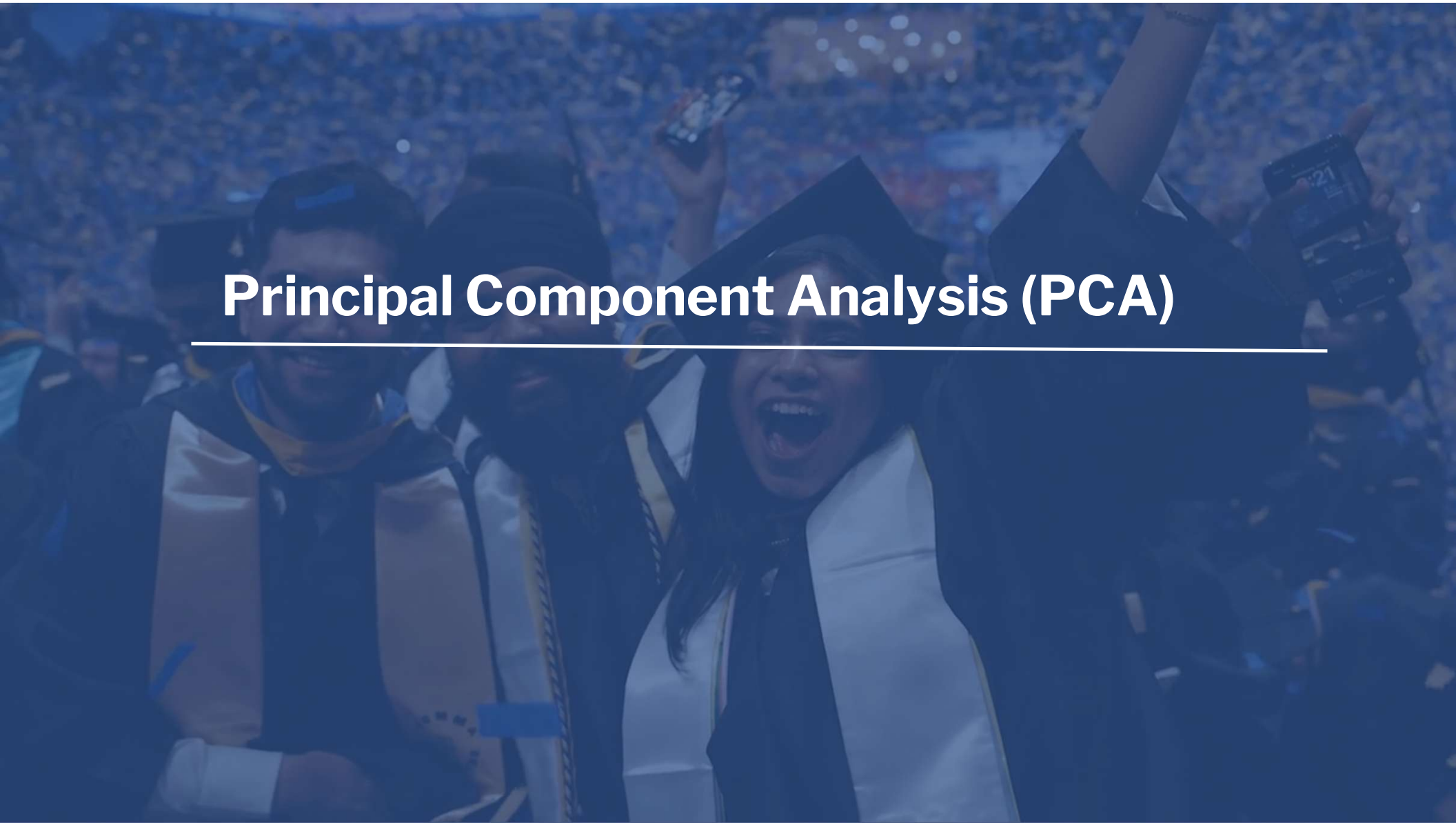
Open Data to Open Opportunities: Building Tools, Research, and Community



Just because data is open doesn't mean it's accessible. While governments make vast amounts of data publicly available, accessing and working with that data can still present significant challenges. How can we break down these barriers and turn open data into meaningful research, tools, and learning opportunities? In this talk, Christian Martinez will introduce the Open Data Lab, an initiative dedicated to making public data more accessible through open-source software, reproducible research, and community collaboration. The presentation will explore a growing ecosystem of R packages designed to simplify access to government open data portals, beginning with nycOpenData and expanding to cities and regions around the United States. Beyond software development, the talk will highlight how the Open Data Lab brings together students and contributors to work on real-world projects, including interactive dashboards, civic data analysis, and open-source tools, while creating opportunities for students to develop technical skills and produce work that extends beyond the classroom.

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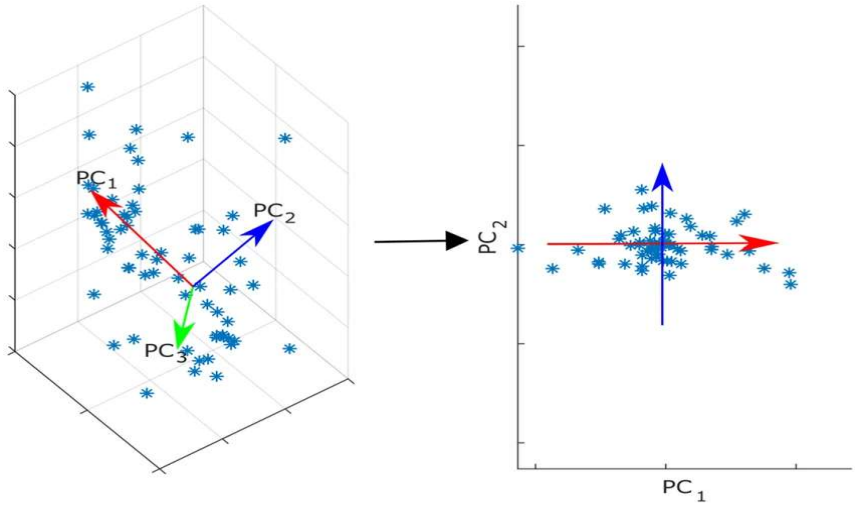
https://us02web.zoom.us/meeting/register/RQW4PGPcT0ixadrHwD0-aQ#/registration



Principal Component Analysis (PCA)

Use-cases for Dimensionality Reduction

- Visualization of high-dimensional data
- Image compression
- Factor analysis
- Reduction in overfitting
- Reduce impact of Curse of Dimensionality
- Reduction in feature correlation
- Reduction in model training time/compute
- Reduction in model complexity
- Transform non-linear data into linearly-separable data
- Reduction in feature noise
- Reduction in neural network complexity (autoencoders)



Source: Towards data science, R. Pramoditha, https://towardsdatascience.com/how-do-you-apply-pca-to-logistic-regression-to-remove-multicollinearity-10b7f8e89f9b / Adaptive dimensionality reduction for neural network-based online principal component analysis, Nico Migenda, et al March 2021

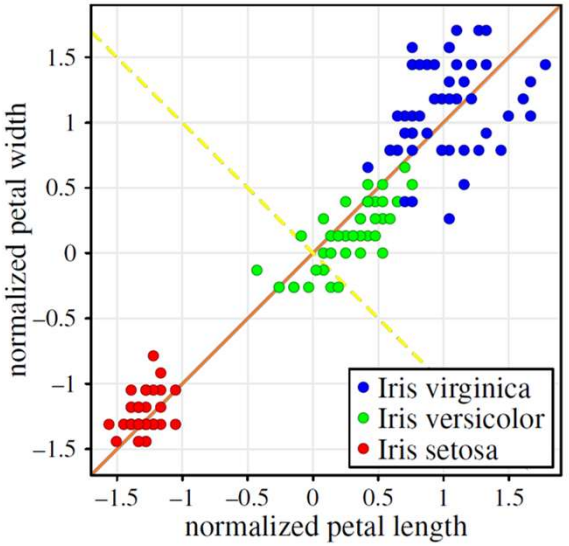
Principal Component Analysis (PCA)

- **Dimensionality reduction technique**
- **Project data from the high-dimensional space to a lower-dimensional space**
- **Criteria: Maximize data variance to construct principal components**

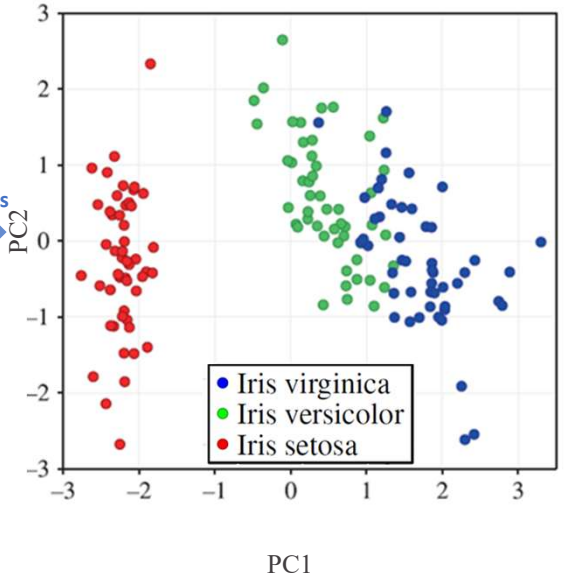
Steps:

1. Construct a line in the direction of maximum variance in data (PC1 = orange line)
2. Next component is orthogonal to the previous component (PC2 = yellow line)

Repeat for as many PC as you need.



Replot onto new components

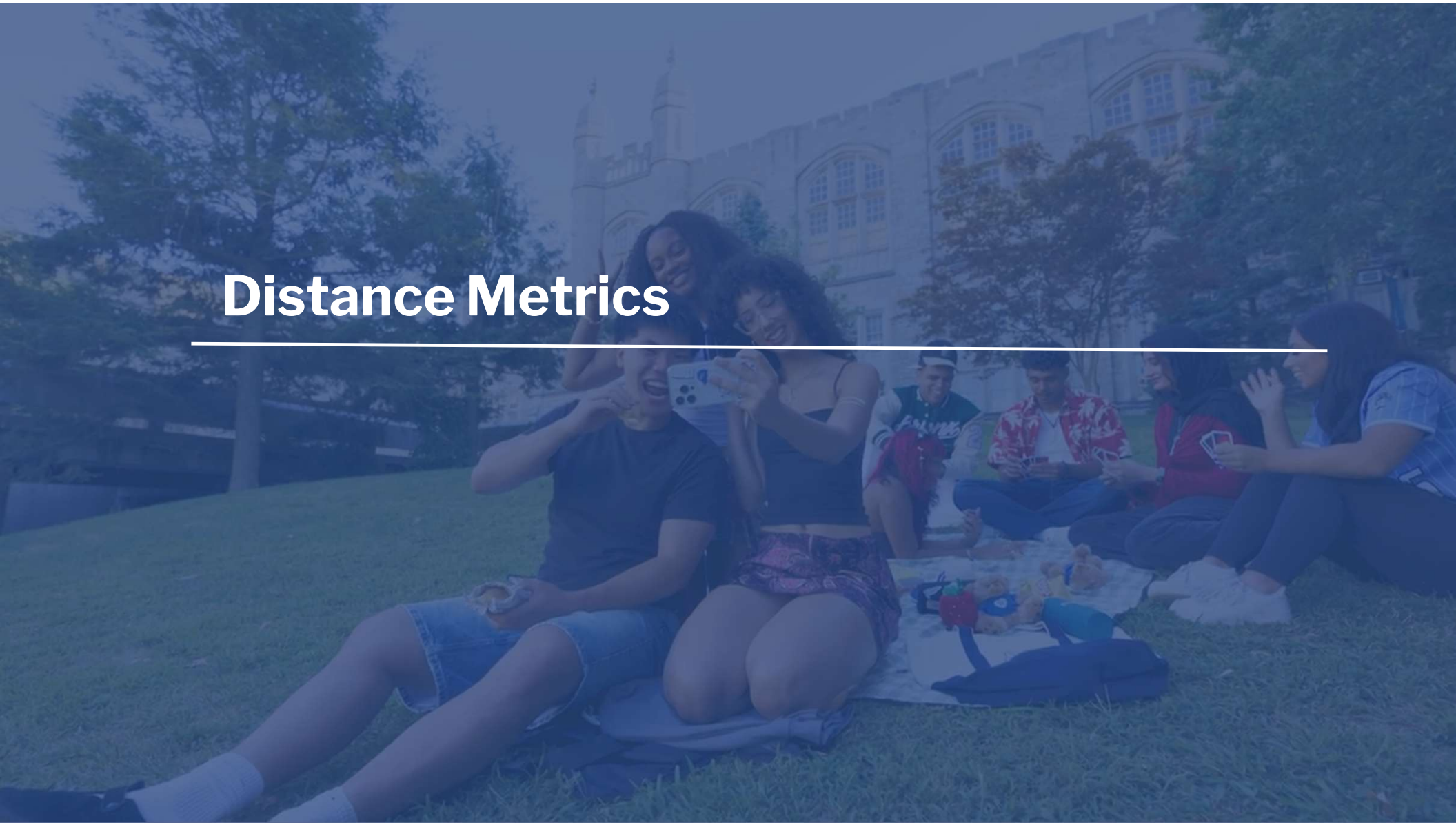


Source: Guide to Intelligent Data Science, Berthold et al

Interpreting PCA results

Business examples:

- Amazon recommending similar items to what you search
- Netflix recommendation of similar movies to what you watched
- Zillow's Zestimate of house price based on similar houses
- Spotify's song/podcast/artist recommendation
- Face recognition on Facebook or screen-lock on cell-phones
- Semantic similarity (e.g., similar words, e.g., Google's dictionary help)
- Facebook's friend recommendation



Distance Metrics

Euclidean Distance

Geometric distance between two points

Also Called: Shortest Path, L2 Distance

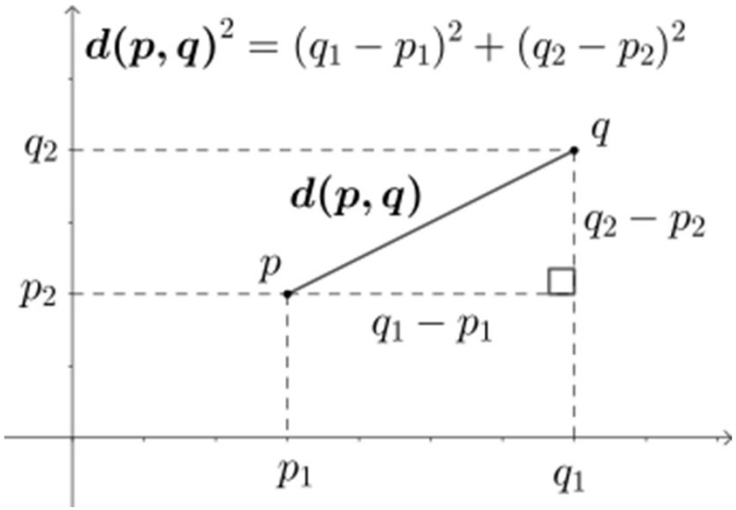
Pros: True distance metric

Cons: Sensitive to both magnitude and direction

Range: [0, ∞)

Used:

- Measuring actual distance is meaningful (e.g., K-means, clustering)
- Working with numerical or spatial features
- You have normalized data



$$D(x,y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2)}$$

Source: Wikipedia

Manhattan Distance

Sum of distances along axes

Also Called: L1 Norm, Taxicab Distance

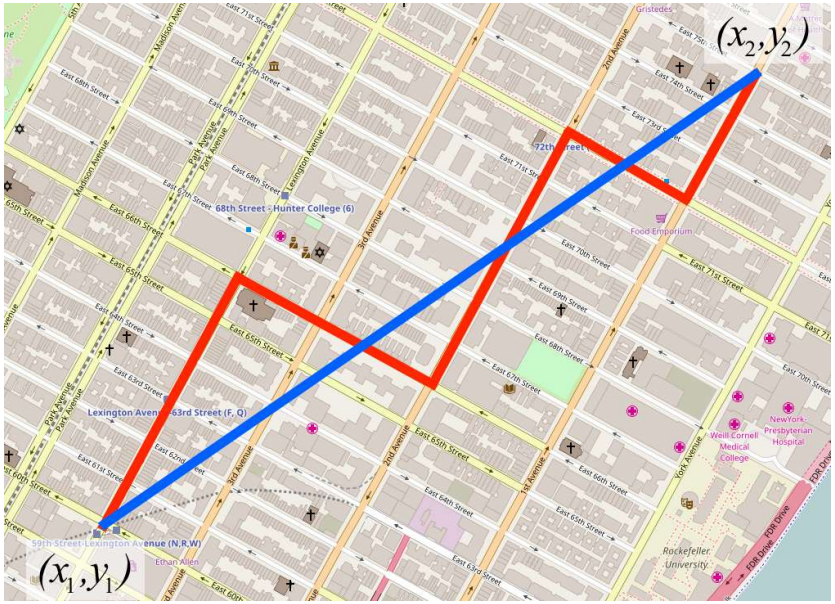
Pros: Simple, more robust to sparse data or outliers

Cons: Low-dimensional data

Range: [0, ∞)

Used:

- Categorical data/entire integers (1,2,3,4…)
- Working with high-dimensional, sparse data (e.g., recommender systems)
- You want interpretable differences in individual features
- Reinforcement learning (discreet steps)



$$Manhattan\,(A, B) = \sum_{i=1}^{N} |fa_i - fb_i|$$

Source: Wikipedia

Cosine Similarity

Similarity between two vectors

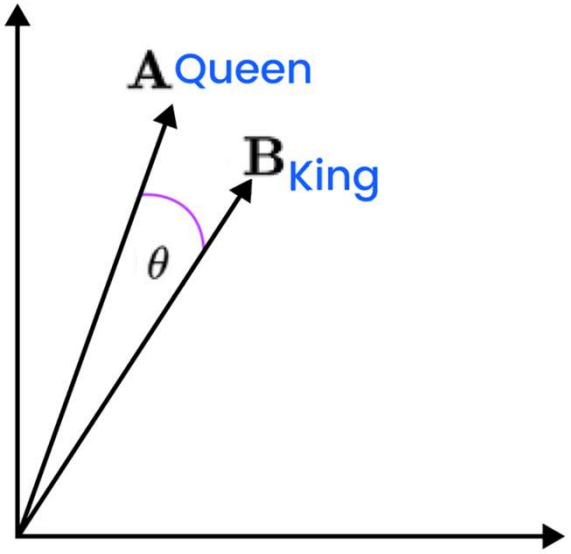
Pros: NLP, robust to high-dimensional or sparse

Cons: Vectors must be normalized

Range: [-1, +1]

Used:

- NLP: Comparing text, documents, or embeddings
- Working with high-dimensional, sparse data
- Magnitude differences are not important



$$\cos(\theta) = \frac{\boldsymbol{A}.\boldsymbol{B}}{\|\boldsymbol{A}\|\|\boldsymbol{B}\|} \sim \boldsymbol{A}.\boldsymbol{B} \quad when\ \|A\| = \|B\| = 1$$

Source: Wikipedia

Dot Product

Generalized form of Cosine similarity

Also called: Inner product

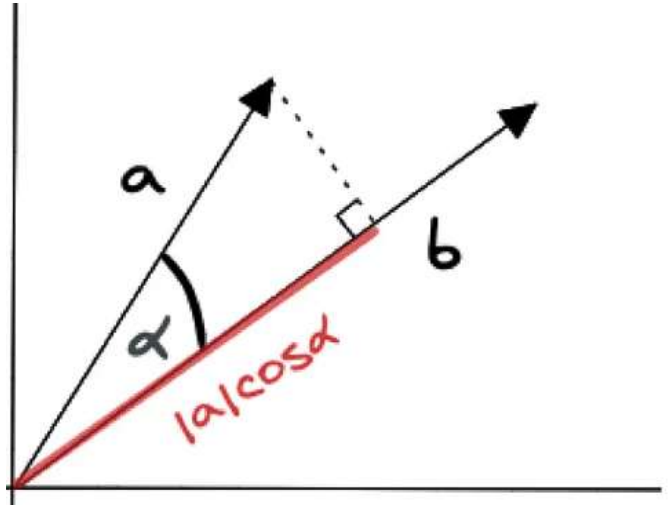
Pros: Not normalize/generalized cosine similarity

Cons: Unbounded, sensitive to vector magnitude

Range: (- ∞, +∞)

Used:

- When magnitude is important (e.g. scores)
- Neural networks and linear models
- Ranking or similarity with unnormalized vectors



$$\boldsymbol{a}.\boldsymbol{b} = \|\boldsymbol{a}\|\|\boldsymbol{b}\|\cos(\boldsymbol{\alpha})$$

Source: Wikipedia

Hamming Distance

Counts the number of positions where two vectors differ

Also called: Hamming Metric

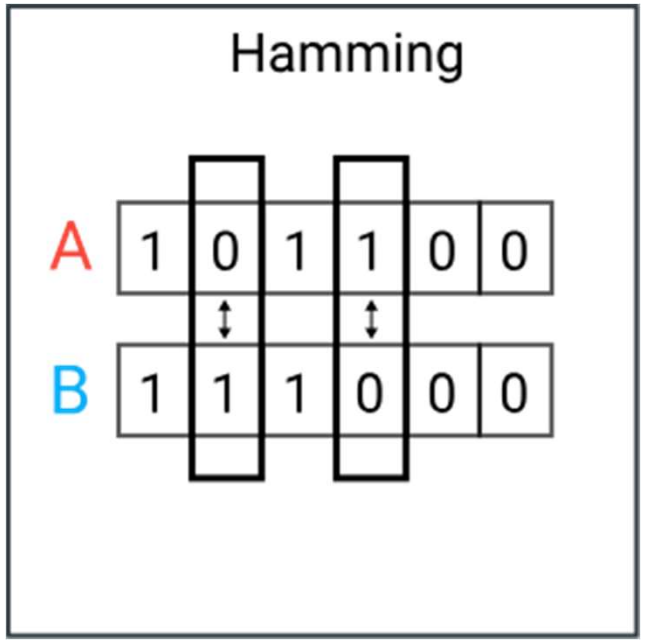
Pros: Fast, binary and categorical data, easy to interpret

Cons: Dimensionality (equal-length vectors), ignores magnitude, not suitable for continuous data

Range: [0-n], n=number of dimensions

Used:

- Comparing binary strings or categorical features
- Error detection/correction
- Feature mismatch analysis in ML



Source: Wikipedia

Sørensen-Dice Index

Measures similarity based on overlap between two datasets

Also called: Dice Coefficient

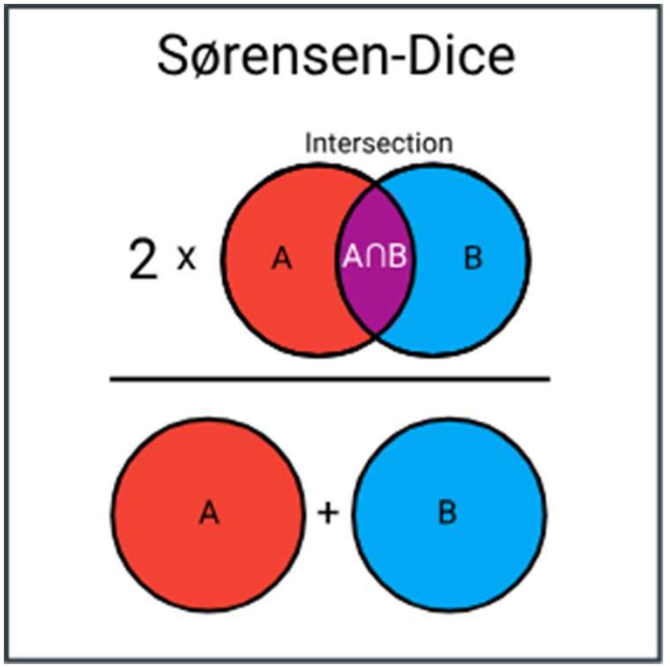
Pros: Emphasizes shared elements (overlap)

Cons: Sensitive to small datasets, less useful for continuous data

Range: [0-1]

Used:

- Text similarity (e.g., n-grams), NLP, and image segmentation (e.g., masks)
- Comparing sets, tokens, or binary features



$$D(x, y) = \frac{2\,|x \cap y|}{|x| + |y|}$$

Source: Wikipedia

Jaccard Distance

Based on non-overlapping elements between two sets

Also called: 1-Jaccard similarity

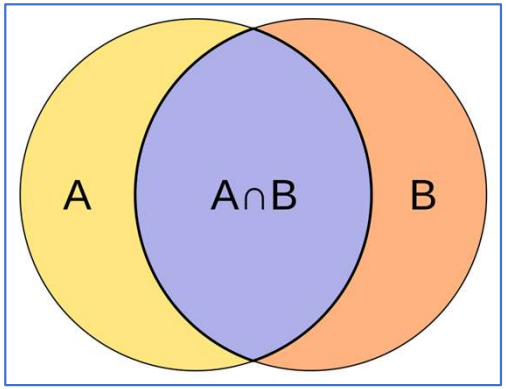
Pros: Handles binary, set data, and sparse data effectively, easy to interpret overlap vs difference

Cons: Ignores frequency/magnitude of features. Not suitable for small datasets and continuous data

Range: [0-1]

Used:

- Text similarity (e.g., word sets)
- Recommender systems with sparse interactions
- Recommendation systems, customer segmentation



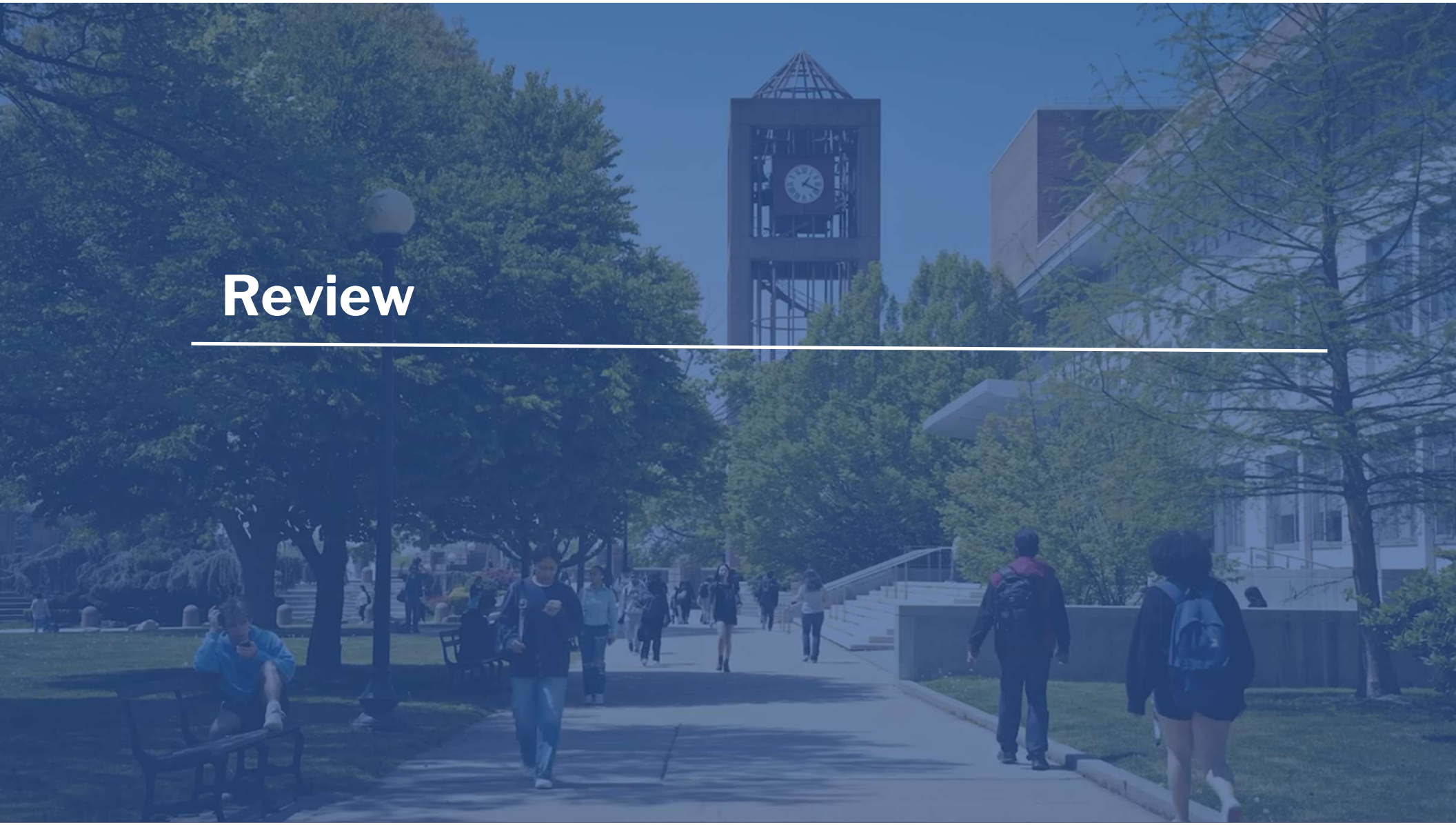
$$J(A,B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}.$$

Source: Wikipedia

Comparison

Distance Metric	Suitable For	Example Applications
Euclidean	Numeric data, clustering	K-Means, PCA
Manhattan	High-dimensional sparse data	Lasso Regression
Minkowski	Flexible generalization	Hybrid distance functions
Cosine	Text similarity	NLP, recommendation systems
Mahalanobis	Correlated data	Anomaly detection
Chebyshev	Maximum absolute difference	Board games
Jaccard	Set comparisons	Text mining, clustering
Hamming	Binary data	Error detection, DNA sequencing
Haversine	Geospatial data	GPS, navigation

Source: Wikipedia

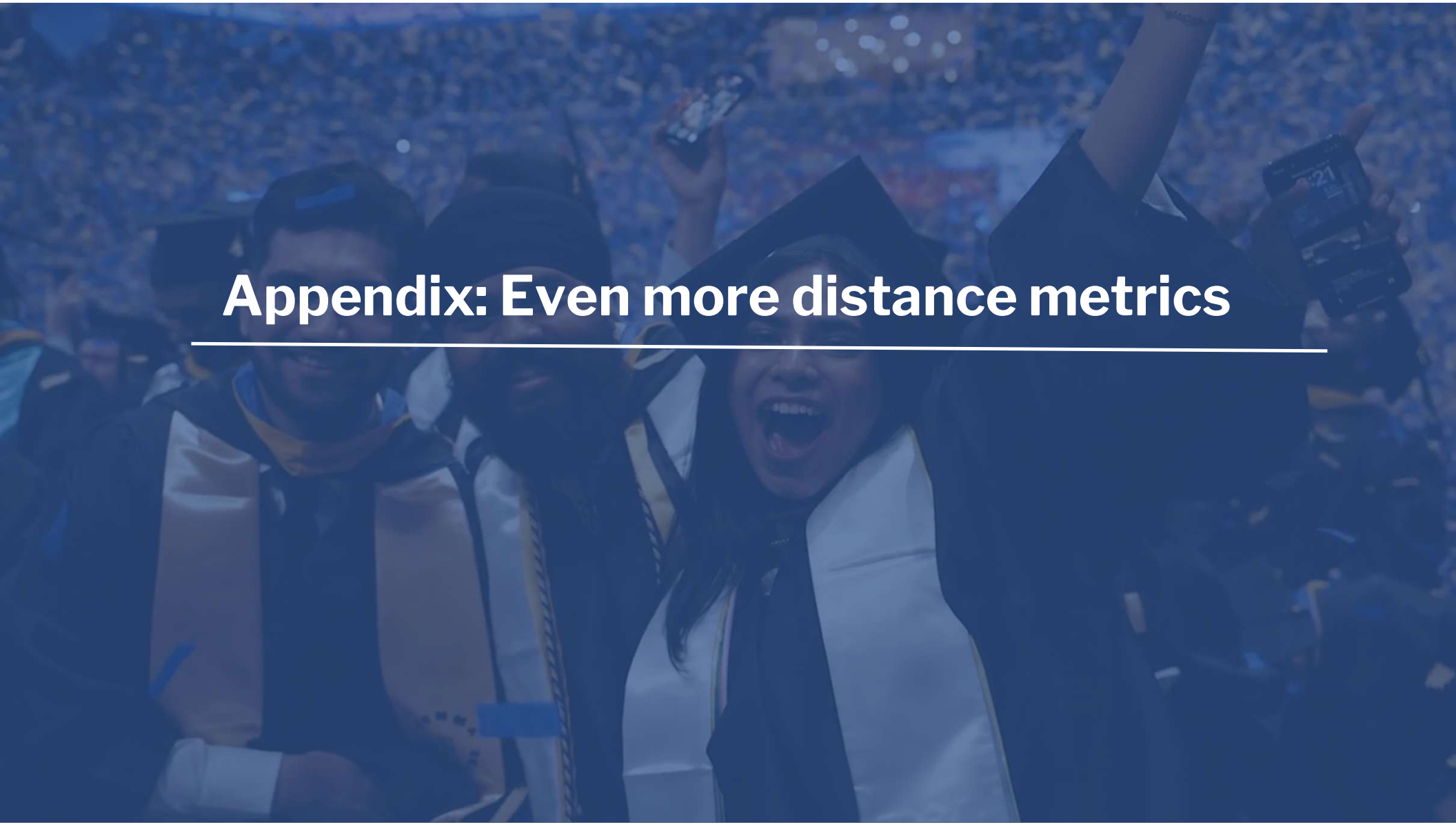


Review

This week we covered

Lesson Objectives/Topics

- **Foundational**
  - Perform dimensionality reduction using PCA for visualization and modeling.
  - Interpret principal components and explain their business implications.
  - Apply distance metrics to assess similarity in multidimensional datasets.
  - Interpret PCA results to inform strategic decision-making with simplified data.



Appendix: Even more distance metrics

Chebyshev Distance

Measures the largest absolute difference between vector elements

Also called: L∞ Norm, Chessboard Distance, Maximum Distance

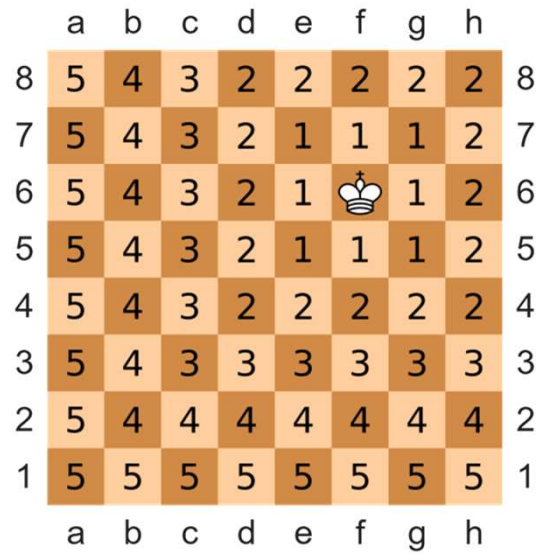
Pros: Simple and efficient to compute, captures worst-case difference across dimensions, useful in grid-based movement
(e.g., Reinforcement Learning use-cases like chess king moves)

Cons: Ignores all but the largest difference, not sensitive to overall variation across dimensions, can be misleading if one feature dominates

Range: [0-∞]

Used:

- Scenarios where the largest deviation matters most
- Grid-based pathfinding (e.g., RL, chessboard, robotics)
- Quality control or tolerance checks (max error threshold)



$$d_C(A, B) = \max_i (|A_i - B_i|)$$

Source: Wikipedia

Haversine Formula

Distance between points on a sphere

Also called: Orthodromic or Great-circle distance

Pros: Measures real-world distances on a sphere

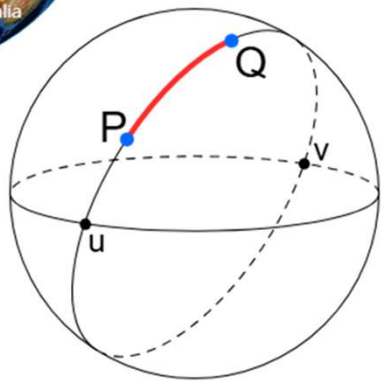
Cons: Assumes point lie on a sphere, more computationally expensive than Euclidian distance

Range: [0-πr]

Used:

- Navigation, calculating distances between GPS coordinates
- Geographic clustering, mapping, or proximity queries





$$d = 2r \arcsin\left(\sqrt{\sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right)}\right)$$

where

- ϕ_1, ϕ_2 = latitudes (in radians)
- λ_1, λ_2 = longitudes (in radians)
- r = radius of the Earth

Source: Wikipedia

Mahalanobis Distance

Measures distance while accounting for correlations among features

Also called: Statistical Distance

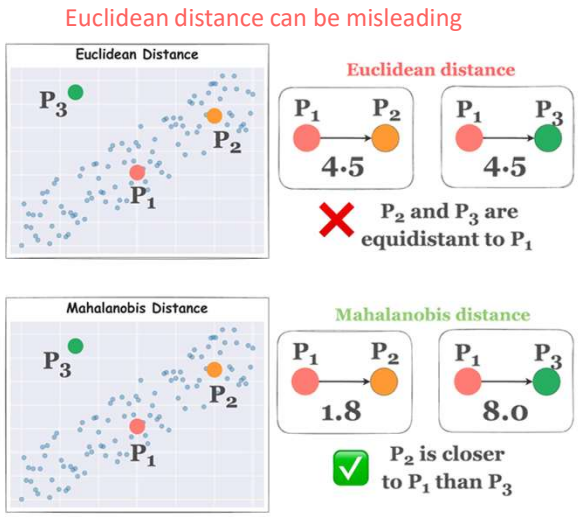
Pros: Accounts for feature correlations, scale-invariant , effective for detecting multivariate outliers

Cons: Requires estimation and inversion of covariance matrix, sensitive to poorly estimated covariance, computationally expensive in high dimensions

Range: [0-∞]

Used:

- Outlier detection in multivariate data
- Anomaly detection (e.g., fraud, sensor data)
- Where feature correlation matters (unlike Euclidean distance)



$$d_M(x, \mu) = \sqrt{(x - \mu)^T \Sigma^{-1} (x - \mu)}$$

where

x = data point

μ = mean vector

Σ = covariance matrix

Source: https://blog.dailydoseofds.com/p/why-prefer-mahalanobis-distance-over

Minkowski Distance

Generalizes distance by combining differences across dimensions using a parameter p

Also called: Lp Norm Distance

Pros: Adjustable sensitivity to large differences, generalization/flexible: unifies multiple distance metrics

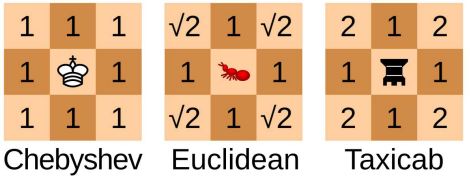
- p=1 → Manhattan distance
- p=2 → Euclidean distance

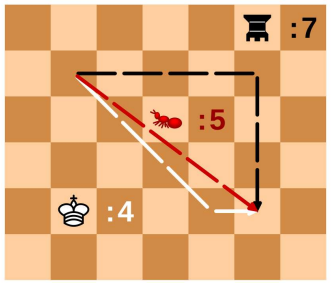
Cons: Choice of p can be arbitrary, higher p values overemphasize large differences, less interpretable

Range: [0-∞]

Used:

- When you want a tunable distance metric
- Experimenting with different norms in ML models
- Situations requiring a balance between L1 and L2 behavior





$$d_M(A,B) = \left(\sum_{i=1}^{n} |A_i - B_i|^p\right)^{1/p}$$

Source: Wikipedia


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